

# Hailey Garcia

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Department of Mathematics, Louisiana State University, Baton Rouge, LA 70803

## Education

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Ph.D. Mathematics, Louisiana State University Fall 2025–present

*Advisor: Christin Bibby*

B.S. Mathematics, Louisiana State University Fall 2022–Spring 2025

## Papers and Preprints

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H. J. Garcia. *Classification of strictly resistance nonnegative graphs*. (2026). [arXiv:2609.23394](https://arxiv.org/abs/2609.23394).

G. Agrahari, C. Bibby, S. Boros, H. J. Garcia, F. Heidercheidt, Z. Wang. *On some structural properties of graphs with non-negative resistance curvature*. (2026). [arXiv:2607.13169](https://arxiv.org/abs/2607.13169). [Code](#).

M. Fairchild, H. J. Garcia, J. Murphy, H. Percle. *On the non-orientable 4-genus of torus knots with braid index 5 or 6*. *Journal of Knot Theory and Its Ramifications* **35** (2026), No. 7, 2650001. [doi:10.1142/S021821652650001X](https://doi.org/10.1142/S021821652650001X).

## Talks

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On some structural properties of graphs with non-negative resistance curvature Fall 2026

*LSU Combinatorics Seminar*

Noncontractive Function Systems Spring 2023

*6th TX-LA Mathematics Undergraduate Conference*

On The “Kelly Sequence” Spring 2023

*6th TX-LA Mathematics Undergraduate Conference*

## Teaching

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MATH 1431, Business Calculus — Lab Leader/TA, LSU Spring 2026  
*Sections 7, 11, 19*

MATH 1431, Business Calculus — Lab Leader/TA, LSU Fall 2025  
*Sections 7, 8, 9, 11, 25, 28*

Honors Calculus 1 — Primary Instructor, Arden Cahill Academy Fall 2023–Spring 2024

## Mentoring

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Mentor, [Directed Reading Program](#) Fall 2026

TA, BSM REU at Haverford College Summer 2026

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|-----------------------------------------------------|-----------------------|
| TA, 2026 LSU Undergraduate FRC                      | Summer 2026           |
| Mentor, Vertically Integrated Research Courses, LSU | Fall 2025–Spring 2026 |

## Organisations

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Organiser, LSU Informal Combinatorics Seminar                                                                                                                          | Fall 2025–present   |
| Webmaster, LSU AMS Student Chapter                                                                                                                                     | Fall 2026           |
| Webmaster, <a href="#">LSU RTG</a>                                                                                                                                     | Spring 2026–present |
| Manager, LSU Math 3D Print Lab                                                                                                                                         | Spring 2026–present |
| Officer, <a href="#">LSU AWM Student Chapter</a><br><i>President (Fall 2025–Spring 2026); Secretary (Spring 2026–present);</i><br><i>Webmaster (Fall 2025–present)</i> | Fall 2025–present   |

## Fellowships and Awards

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|--------------------------------------------------------------|-------------------|
| LSU RTG Fellow                                               | Fall 2026–present |
| Pasquale Porcelli Graduate Student Academic Excellence Award | Spring 2026       |
| Robert and Betti Giles Senior Mathematics Award              | Spring 2025       |
| Peg and Tom Madden Mathematics Research Scholarship          | Spring 2024       |
| Porcelli Junior Scholarship                                  | Spring 2023       |